CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Data Analytics Project Report

Python | SQL Server 2022 | Power BI

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Date: October 2026

Executive Summary

This project analyses customer shopping behavior using Python for data preparation, SQL Server 2022 for structured business analysis, and Power BI for interactive visualization. The dataset contains 3,900 shopping records and 18 source columns, including customer demographics, product category, purchase amount, review rating, shipping type, discount usage, subscription status, and previous purchases. Ten SQL business questions were used to examine revenue, product ratings, discounts, shipping, customer segments, repeat buyers, and age-group revenue.

The supplied dashboard screenshot displays 648 customers, an average purchase amount of $58.63, and an average review rating of 3.72. The Shipping Type slicer visibly has Next Day Air selected; therefore, these KPI values represent the selected filter context and should not be treated as unfiltered dataset totals.

1. Project Overview

The purpose of the project is to understand customer purchasing patterns and provide useful insights for retail decision-making. The analysis compares spending across demographic groups, identifies high-rated and frequently purchased products, studies discount usage, and examines subscription and repeat-purchase behavior.

2. Dataset Summary

Attribute	Description
Dataset file	customer_shopping_behavior.csv
Rows	3,900
Source columns	18
Missing values	37 missing values in Review Rating in the raw dataset
Key fields	Age, Gender, Item Purchased, Category, Purchase Amount (USD), Review Rating, Subscription Status, Shipping Type, Discount Applied, Previous Purchases

The data contains demographic, product, purchase, shipping, discount, subscription, and purchase-frequency attributes. Each row is treated as a shopping record. Customer ID is available as an identifier.

3. Tools and Technologies

Tool	Purpose
Python / Pandas	Data loading, inspection, cleaning, and feature engineering.
SQL Server 2022 / SSMS	Store the cleaned data and run SQL business queries.
SQLAlchemy + pyodbc	Connect Python to SQL Server and transfer the DataFrame.
Power BI Desktop	Create KPI cards, charts, and interactive slicers.
Jupyter Notebook / VS Code	Write and execute the Python workflow.

4. Python Data Preparation

- Loaded the CSV file into a Pandas DataFrame.
- Used df.info(), df.describe(), and missing-value checks to inspect data structure and summary statistics.
- Handled missing Review Rating values using the median rating within each product category, as described in the reference workflow.
- Standardized column names to lowercase snake_case for easier use in SQL.
- Created age_group using age-based bins and created purchase_frequency_days from purchase frequency labels.
- Checked whether Discount Applied and Promo Code Used were redundant before removing a duplicate field.

Note: The report describes the preparation workflow associated with the project. The final cleaned DataFrame and exact Python output were not supplied as screenshots in this conversation.

5. SQL Server 2022 Analysis

The prepared data is stored in the customer_behavior database in SQL Server 2022. The SQL queries use the dbo.customer table and SQL Server syntax such as USE, GO, TOP, CAST, and DECIMAL. The following sections include the business question, the SQL used, the cropped output from the screenshots provided, and a short interpretation.

Q1. Revenue by Gender

What is the total purchase revenue generated by male and female customers?

SQL Query

```
SELECT gender,
    CAST(SUM(purchase_amount) AS DECIMAL(10,2)) AS
revenue
FROM dbo.customer
GROUP BY gender;
```

SQL Server Output

gender	revenue
Male	157890.00
Female	75191.00

Interpretation: Male records contribute 157,890.00 in total purchase amount, compared with 75,191.00 for female records. This is total recorded revenue, not average spend per customer.

Q2. Discount Users with High Spending

Which customers used a discount and spent at least the overall average purchase amount?

SQL Query

```
SELECT customer_id, purchase_amount
FROM dbo.customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (
    SELECT AVG(purchase_amount) FROM dbo.customer
  );
```

SQL Server Output

customer_id	purchase_amount
2	64.00
3	73.00
4	90.00
7	85.00
9	97.00
12	68.00
13	72.00
16	81.00

Interpretation: The output lists customers who meet both conditions. The screenshot shows the first eight rows of a larger result; the full result count is not visible in the supplied image.

Q3. Top 5 Products by Average Rating

Which five products have the highest average review rating?

SQL Query

```
SELECT TOP 5 item_purchased,
    CAST(ROUND(AVG(CAST(review_rating AS
DECIMAL(10,2))),2)
    AS DECIMAL(10,2)) AS [Average Product Rating]
FROM dbo.customer
GROUP BY item_purchased
ORDER BY AVG(CAST(review_rating AS DECIMAL(10,2))) DESC;
```

SQL Server Output

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Interpretation: Gloves rank first with an average rating of 3.86, followed by Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78).

Q4. Average Purchase by Shipping Type

How do average purchase amounts compare for Standard and Express shipping?

SQL Query

```
SELECT shipping_type,
    CAST(ROUND(AVG(CAST(purchase_amount AS
DECIMAL(10,2))),2)
    AS DECIMAL(10,2)) AS avg_purchase_amount
FROM dbo.customer
WHERE shipping_type IN ('Standard','Express')
GROUP BY shipping_type;
```

SQL Server Output

shipping_type	avg_purchase_am
Standard	58.46
Express	60.48

Interpretation: Express shipping has an average purchase amount of 60.48, compared with 58.46 for Standard shipping.

Q5. Subscribers vs. Non-Subscribers

Compare customer counts, average purchase amount, and total revenue by subscription status.

SQL Query

```
SELECT subscription_status,
    COUNT(customer_id) AS total_customers,
    CAST(ROUND(AVG(CAST(purchase_amount AS DECIMAL(10,2))),2)
    AS DECIMAL(10,2)) AS avg_spend,
    CAST(SUM(purchase_amount) AS DECIMAL(10,2)) AS total_revenue
FROM dbo.customer
GROUP BY subscription_status
ORDER BY total_revenue DESC, avg_spend DESC;
```

SQL Server Output

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436.00
Yes	1053	59.49	62645.00

Interpretation: Non-subscribers account for 2,847 records and 170,436.00 in total revenue; subscribers account for 1,053 records and 62,645.00. Average spend is 59.87 for non-subscribers and 59.49 for subscribers.

Q6. Products with Highest Discount Rate

Which five products have the highest percentage of purchases with discounts applied?

SQL Query

```
SELECT TOP 5 item_purchased,
    CAST(ROUND(100.0 * SUM(CASE WHEN
discount_applied = 'Yes'
    THEN 1 ELSE 0 END) / NULLIF(COUNT(*),0),2)
    AS DECIMAL(10,2)) AS discount_rate
FROM dbo.customer
GROUP BY item_purchased
ORDER BY discount_rate DESC, item_purchased;
```

SQL Server Output

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Interpretation: Hat has the highest discount rate at 50.00%, followed by Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%).

Q7. Customer Segmentation

How many customers fall into New, Returning, and Loyal segments based on previous purchases?

SQL Query

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
      CASE WHEN previous_purchases <= 1 THEN 'New'
          WHEN previous_purchases BETWEEN 2 AND 10
THEN 'Returning'
          ELSE 'Loyal' END AS customer_segment
  FROM dbo.customer
)
SELECT customer_segment, COUNT(*) AS [Number of
Customers]
FROM customer_type
GROUP BY customer_segment;
```

SQL Server Output

customer_segment	Number of Cust
New	83
Returning	701
Loyal	3116

Interpretation: The output shows 83 New, 701 Returning, and 3,116 Loyal records. In this SQL logic, previous_purchases values of zero or one are both classified as New.

Q8. Top Products Within Each Category

Which three products are most purchased within each category?

SQL Query

```
WITH item_counts AS (
  SELECT category, item_purchased,
      COUNT(customer_id) AS total_orders,
      ROW_NUMBER() OVER (
        PARTITION BY category
        ORDER BY COUNT(customer_id) DESC,
item_purchased
      ) AS item_rank
  FROM dbo.customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased,
total_orders
FROM item_counts
WHERE item_rank <= 3
ORDER BY category, item_rank;
```

SQL Server Output

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150

Interpretation: The visible output ranks products within each category. Accessories: Jewelry leads with 171, followed by Belt and Sunglasses at 161. Clothing: Blouse and Pants each show 171, followed by Shirt at 169. The supplied screenshot cuts off some later category rows.

Q9. Repeat Buyers by Subscription Status

Among customers with more than five previous purchases, how many are subscribers and non-subscribers?

SQL Query

```
SELECT subscription_status,
    COUNT(customer_id) AS repeat_buyers
FROM dbo.customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

SQL Server Output

subscription_status	repeat_buyers
No	2518
Yes	958

Interpretation: Among records with more than five previous purchases, 2,518 are marked No and 958 are marked Yes for subscription status. These are counts by status, not subscription probabilities.

Q10. Revenue by Age Group

What is the total purchase revenue contributed by each age group?

SQL Query

```
SELECT age_group,
    CAST(SUM(purchase_amount) AS DECIMAL(10,2)) AS
total_revenue
FROM dbo.customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

SQL Server Output

	age_group	total_revenue
1	Young Adult	62143.00
2	Middle-aged	59197.00
3	Adult	55978.00
4	Senior	55763.00

Interpretation: Young Adult has the highest total at 62,143.00, followed by Middle-aged at 59,197.00, Adult at 55,978.00, and Senior at 55,763.00.

6. Power BI Dashboard

The final Power BI dashboard includes KPI cards for number of customers, average purchase amount, and average review rating. It also contains a subscription-status doughnut chart, revenue and sales by category, revenue and sales by age group, and slicers for subscription status, gender, category, and shipping type.

	age_group	total_revenue
1	Young Adult	62143.00
2	Middle-aged	59197.00
3	Adult	55978.00
4	Senior	55763.00

Figure 1. Final Customer Behavior Dashboard supplied by the project owner.

The screenshot shows 648 customers, average purchase amount of $58.63, and average review rating of 3.72. The Shipping Type slicer is set to Next Day Air, so the KPIs reflect the current filter selection.

7. Key Findings

- Male records have higher total purchase amount than female records in the supplied SQL output.
- Express shipping has a higher average purchase amount than Standard shipping in the displayed comparison.
- Gloves have the highest average review rating among the five listed products.
- Non-subscribers contribute higher total purchase amount than subscribers, while average spend is close between the two groups.
- Hat has the highest discount-use rate among the five products listed.
- The Loyal segment is the largest group under the project's previous-purchase classification.
- Young Adult has the highest total purchase amount among the four age groups.

8. Business Recommendations

- Review Clothing and Accessories performance regularly, as the dashboard indicates they are leading categories by revenue and sales.
- Test subscription offers with clearly defined experiments; compare conversion, repeat purchases, and average spend before concluding impact.
- Evaluate discount performance by product alongside sales volume and profit margin, because discount rate alone does not show profitability.

- Compare shipping options using customer satisfaction, delivery costs, and conversion as well as average purchase amount.
- Use age-group and category slicers to explore differences, while treating quartile-based age groups as dataset-specific segments.
- Keep metric definitions consistent: record counts, unique customers, total purchase amount, and profit are different measures.

9. Limitations

- The source dataset does not provide transaction dates, product costs, profit margins, or actual delivery times.
- The supplied Q2 and Q8 screenshots show only part of their result grids; the report interprets only the visible rows.
- Age groups are derived bins and depend on the project's chosen binning logic.
- Median imputation of review ratings can affect rating variability and should be documented in the notebook.
- Descriptive comparisons do not establish that discounts, shipping, or subscriptions cause higher spending.
- Dashboard KPI values depend on slicer selections and may differ from full-dataset totals.

10. Conclusion

This project demonstrates an end-to-end customer shopping behavior analysis workflow using Python, SQL Server 2022, and Power BI. Python supports data inspection and preparation, SQL Server answers ten business questions, and Power BI presents the results through an interactive dashboard. The analysis identifies differences in total purchase amount by gender, product rating leaders, shipping comparisons, discount-use rates, customer segments, repeat-buyer status, and age-group contribution. Future improvements could include transaction dates, profitability data, cohort analysis, and automated dashboard refresh.

Appendix: Reproducibility Checklist

- Keep the source CSV and Python notebook together.
- Run the Python workflow from top to bottom and verify the cleaned DataFrame.
- Confirm the SQL Server instance, database, table, and column names before running queries.
- Run each SQL query in SSMS and compare results with the report screenshots.
- Refresh the Power BI report and test all slicers before presenting.